

Supplementary Material for NETSHIELD-IDS: Shared-Bottleneck Evidence Scheduling for Multi-Probe IoT Edge Intrusion Detection

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Frozen REV2 analysis baseline: `v0.4.7-analysis-rev2-session-level`

1 Scope and Statistical Unit

This supplement contains statistical and provenance detail omitted from the main manuscript for readability. It does not introduce new experimental data or recompute the frozen empirical results. The formal campaign contains 12 paper-eligible sessions (S01–S12), three capacity regimes, six methods, 216 method–capacity arms, 648 device-runs, and 58,320 measured probe-windows. Each method contributes 108 device-runs and 9,720 measured windows.

The inferential unit is the *session* ($n = 12$). Within each session and method, the three capacities and three probes are pooled before paired method differences are formed. The 36 session–capacity cells are descriptive, not independent inferential replicates. For each metric family, all 15 pairwise method contrasts are tested by exact two-sided sign flip over $2^{12} = 4096$ assignments, with 10,000-resample session-cluster bootstrap confidence intervals and Holm correction across the 15 contrasts in that family.

2 Complete Session-Level Pairwise Contrasts

Table 1 gives all 75 frozen comparisons (five metric families $\times$ 15 method pairs). Method IDs are A=fixed-local, B=fixed-summary, C=fixed-full, D=independent-DPP, E=central-DPP, and F=shielded-DPP. Differences are the first ID minus the second ID. The values are copied directly from `tableS1_full_contrasts.csv` in the frozen paper package.

Table 1: Complete frozen REV2 session-level contrast family.

Metric	Pair	Δ	95% CI	raw p	Holm p	n
FPR	E – C	-0.28172	[-0.29621, -0.26799]	0.000488	0.0073	12
FPR	E – A	0	[0, 0]	1.0000	1.0000	12
FPR	E – B	0	[0, 0]	1.0000	1.0000	12
FPR	E – D	0	[0, 0]	1.0000	1.0000	12
FPR	E – F	0	[0, 0]	1.0000	1.0000	12
FPR	C – A	0.28172	[0.26799, 0.29621]	0.000488	0.0073	12
FPR	C – B	0.28172	[0.26799, 0.29621]	0.000488	0.0073	12
FPR	C – D	0.28172	[0.26799, 0.29621]	0.000488	0.0073	12
FPR	C – F	0.28172	[0.26799, 0.29621]	0.000488	0.0073	12
FPR	A – B	0	[0, 0]	1.0000	1.0000	12
FPR	A – D	0	[0, 0]	1.0000	1.0000	12
FPR	A – F	0	[0, 0]	1.0000	1.0000	12
FPR	B – D	0	[0, 0]	1.0000	1.0000	12
FPR	B – F	0	[0, 0]	1.0000	1.0000	12
FPR	D – F	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	E – C	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	E – A	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	E – B	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	E – D	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	E – F	-0.00010	[-0.00031, 0]	1.0000	1.0000	12
deadline-violation rate	C – A	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	C – B	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	C – D	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	C – F	-0.00010	[-0.00031, 0]	1.0000	1.0000	12
deadline-violation rate	A – B	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	A – D	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	A – F	-0.00010	[-0.00031, 0]	1.0000	1.0000	12
deadline-violation rate	B – D	0	[0, 0]	1.0000	1.0000	12
deadline-violation rate	B – F	-0.00010	[-0.00031, 0]	1.0000	1.0000	12
deadline-violation rate	D – F	-0.00010	[-0.00031, 0]	1.0000	1.0000	12
mean E2E (ms)	E – C	-22.39363	[-40.63494, -4.42483]	0.0420	0.5459	12
mean E2E (ms)	E – A	-2.05747	[-14.02703, 8.47212]	0.7363	1.0000	12
mean E2E (ms)	E – B	-2.26766	[-16.02513, 9.97172]	0.7480	1.0000	12
mean E2E (ms)	E – D	10.75696	[-6.62047, 31.97360]	0.3682	1.0000	12
mean E2E (ms)	E – F	5.81350	[-7.87680, 20.84748]	0.4902	1.0000	12
mean E2E (ms)	C – A	20.33616	[-1.38858, 35.19507]	0.0684	0.8203	12

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Table 1 – continued

Metric	Pair	Δ	95% CI	raw p	Holm p	n
mean E2E (ms)	C – B	20.12597	[-2.75963, 34.14628]	0.0845	0.9292	12
mean E2E (ms)	C – D	33.15059	[19.80053, 48.33126]	0.000977	0.0146	12
mean E2E (ms)	C – F	28.20713	[19.19397, 36.56299]	0.0015	0.0205	12
mean E2E (ms)	A – B	-0.21019	[-5.39903, 5.17789]	0.9390	1.0000	12
mean E2E (ms)	A – D	12.81444	[-6.35393, 37.15630]	0.3838	1.0000	12
mean E2E (ms)	A – F	7.87097	[-5.83860, 28.06705]	0.5864	1.0000	12
mean E2E (ms)	B – D	13.02462	[-6.35064, 38.58680]	0.3613	1.0000	12
mean E2E (ms)	B – F	8.08116	[-7.09734, 30.48696]	0.6646	1.0000	12
mean E2E (ms)	D – F	-4.94346	[-24.36588, 8.97751]	0.7207	1.0000	12
queue-violation rate	E – C	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	E – A	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	E – B	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	E – D	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	E – F	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	C – A	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	C – B	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	C – D	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	C – F	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	A – B	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	A – D	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	A – F	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	B – D	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	B – F	0	[0, 0]	1.0000	1.0000	12
queue-violation rate	D – F	0	[0, 0]	1.0000	1.0000	12
timely-F1	E – C	0.06649	[0.06264, 0.07009]	0.000488	0.0073	12
timely-F1	E – A	0	[0, 0]	1.0000	1.0000	12
timely-F1	E – B	0	[0, 0]	1.0000	1.0000	12
timely-F1	E – D	0	[0, 0]	1.0000	1.0000	12
timely-F1	E – F	0	[0, 0]	1.0000	1.0000	12
timely-F1	C – A	-0.06649	[-0.07008, -0.06264]	0.000488	0.0073	12
timely-F1	C – B	-0.06649	[-0.07008, -0.06264]	0.000488	0.0073	12
timely-F1	C – D	-0.06649	[-0.07008, -0.06264]	0.000488	0.0073	12
timely-F1	C – F	-0.06649	[-0.07008, -0.06264]	0.000488	0.0073	12
timely-F1	A – B	0	[0, 0]	1.0000	1.0000	12
timely-F1	A – D	0	[0, 0]	1.0000	1.0000	12
timely-F1	A – F	0	[0, 0]	1.0000	1.0000	12
timely-F1	B – D	0	[0, 0]	1.0000	1.0000	12
timely-F1	B – F	0	[0, 0]	1.0000	1.0000	12
timely-F1	D – F	0	[0, 0]	1.0000	1.0000	12

3 Non-Inferiority and Action Consistency

The prespecified non-inferiority comparison is shielded-DPP (F) versus fixed-summary (B) in timely F1. The frozen result is $F - B = 0.000$ with margin -0.02 , $n = 12$ sessions, exact sign-flip procedure, and `meets_non_inferiority=true`. The Holm-adjusted equality-test p -value is 1.0; the non-inferiority conclusion follows from the prespecified margin and observed difference rather than from rejecting an equality null.

Table 2: Measured-window action consistency.

Comparison	Windows	Action differences	Interpretation
E vs F	9,720	0	shield non-interference in evaluated envelope
D vs F	9,720	5,940	independent vs shared pre-admission differs by design

The four dominant D→F transition pairs are H-NONE→H-SUMMARY (2,736 windows), H-FULL→H-SUMMARY (1,872), M-SUMMARY→H-SUMMARY (792), and H-NONE→H-FULL (540). Measured-window generated payload is 747,307,008 B for D and 654,704,640 B for F/E; D therefore generates 14.1% more measured-window application payload.

4 Capacity-Specific Fixed-Full Diagnostic

The capacity interaction is descriptive and uses pooled confusion matrices within each capacity. Fixed-full (C) is the only method that loses timely detection quality under low/mid capacity; all other methods remain at F1 1.0000 and FPR 0.0000 in all three capacity regimes.

Table 3: Fixed-full (C) capacity-specific descriptive diagnostics.

Capacity	timely F1	FPR	TP/FP/FN/TN	age > 10 windows	rate
Low	0.9375	0.2797	2452/179/148/461	2916/3240	90.0%
Mid	0.8634	0.5672	2251/363/349/277	2376/3240	73.3%
High	1.0000	0.0000	2600/0/0/640	0/3240	0.0%

These age counts are a separate descriptive diagnostic from the configured primary queue-violation indicator. A zero primary queue-violation rate therefore must not be read as zero backlog or zero stale evidence.

5 Unique Shielded-DPP Deadline Miss

F records one measured deadline violation in 9,720 windows, giving the Table-1 run-macro violation rate 0.000103. The event is S09/high/P3/window 80: decision latency 1315.090898 ms against the 1000-ms deadline, evidence completion 1578.689249 ms, and evidence receipt RTT 1311.904469 ms. Queue bytes are zero, the action is H-FULL, and neighboring RTT values are approximately 66/63 ms. This is retained as evidence that configured byte feasibility is not an unconditional guarantee on stochastic Wi-Fi completion latency.

6 Primary Table Aggregation Check

Table-1 timely-F1/FPR/E2E/violation values are unweighted macro means over the 108 device-runs per method. For fixed-full, the run-macro values are timely F1 0.933168 and FPR 0.284536 (reported 0.9332/0.2845). Pooling all window-level confusion counts instead gives micro F1 0.933589 and FPR 0.282292 (reported only as a reference). MaxQ is measured in MiB: C reaches 8,301,290 B = 7.92 MiB and D reaches 87,382 B = 0.08 MiB.

7 Artifact Identity

- Statistical truth: commit `a24b909`, tag `v0.4.7-analysis-rev2-session-level`.
- Paper-facing documentation: commit `92c3108`, tag `v0.4.7-analysis-rev2-doc-clean`.
- Clean data-package SHA-256: `561742f99e7b4d389fd2077cf6485c04c432468ce692ef8c54144525ecfd8092`.
- The package manifest verifies 18/18 hashed entries with `shasum -a 256 -c`.